Bonn International School

Chemistry

Standard Level
Paper 2

Wednesday 28 May 2025

Instructions to candidates

- Do not open this examination paper until instructed to do so.
- Answer all the questions. Write your answers in the boxes provided.
- A clean copy of the Chemistry data booklet is required for this paper. A calculator is also permitted for this paper.
- Each question has a different number of marks. The maximum mark for Paper 2 is **[50 marks].**

12 Pages

1. Potassium, K, and potassium chloride, KCl, both form lattice structures in the solid state.

(a) Predict, with a reason, the electrical conductivity of K (s) and KCl (s). [2]

K (s): ....

....

....

KCl (s): ....

....

....

(b) State the number of each type of subatomic particle in the potassium ion, $^{41}_{19}K^{+}$. [1]

Protons: ....

Electrons: ....

Neutrons: ....

(c) Potassium reacts with water to produce potassium hydroxide.

$$2K\,(s) + 2H_2O\,(l) \rightarrow 2KOH\,(aq) + H_2\,(g)$$

Describe the difference between the reactions of sodium and potassium with water. [1]

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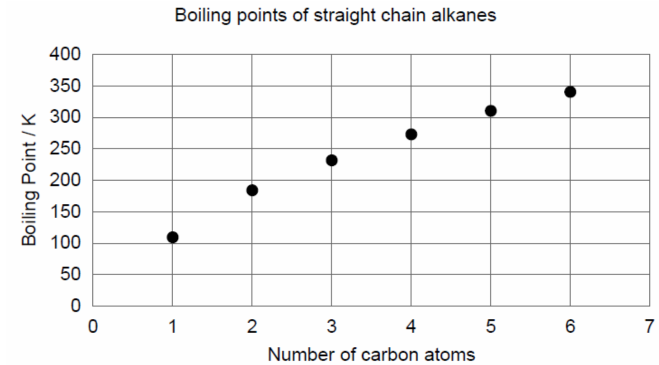
2. Alkanes are commonly occurring organic compounds.

(a) The first four straight chain alkanes are gases at room temperature.



(a.i) Explain why the boiling point increases from methane to propane. [2]

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(a.ii) Explain why the volume occupied by a sample of propane increases sharply when the sample is heated up from 200 to 250 K at constant pressure. [2]

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(a.iii) *There is no question a(iii).*

(a.iv) Outline why the volume occupied by propane (g) at very high pressure is higher than the value calculated using PV = nRT. [2]

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3. NO_2 is a brown gas found in photochemical smog. NO_2 forms from the reaction of NO and O_2 in the atmosphere.

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

(a) Suggest a method to measure the rate of reaction. [1]

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4. The Winkler Method is used to find the concentration of oxygen in water.

200 cm^3 of water was taken from a river and analysed using this method. The reactions taking place are summarized.

Step 1: $2Mn^{2+}(aq) + 4OH^-(aq) + O_2(aq) \rightarrow 2MnO_2(s) + 2H_2O(l)$

Step 2: $MnO_2(s) + 2I^-(aq) + 4H^+(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$

Step 3: $2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

(a) There is no question a.

(b) 4.8×10^{-3} moles of I^- were formed in step 3. Determine the number of moles of O_2 in the water sample. [1]

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(c) Calculate the concentration of oxygen, in $g\,dm^{-3}$, in the water sample. If you did not obtain an answer in (a), use 2.0×10^{-3} mol although this is not the correct answer. [1]

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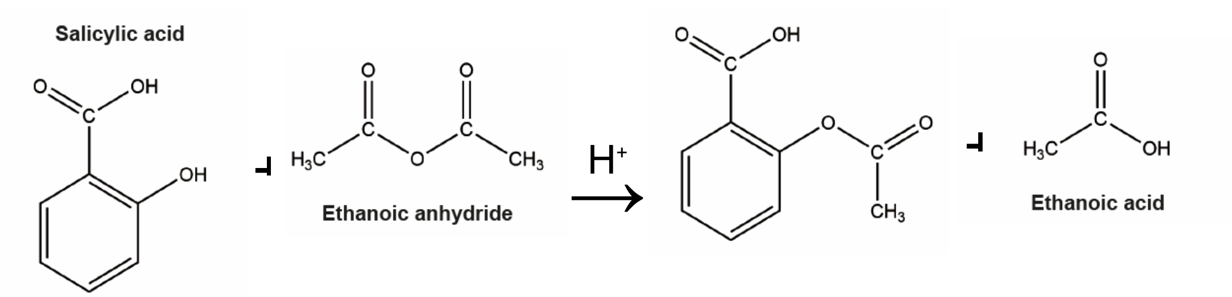
5. Aspirin was synthesized by mixing 0.897 g of salicylic acid with excess ethanoic anhydride.

M_r (salicylic acid) = 138.13. M_r (aspirin) = 180.17



(a.i) Calculate the theoretical yield of aspirin, in g. [1]

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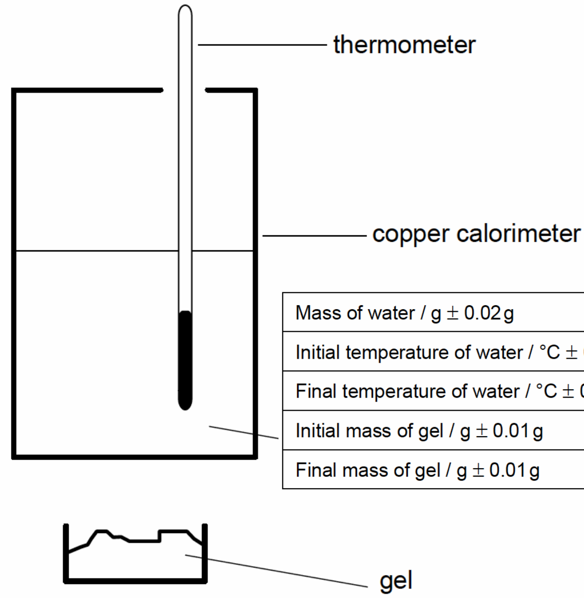
6. A student investigated the use of hand sanitising gel containing propan-1-ol as a camping fuel.



Mass of water / g ± 0.02 g	400.00
Initial temperature of water / °C ± 0.5 °C	19.0
Final temperature of water / °C ± 0.5 °C	40.0
Initial mass of gel / g ± 0.01 g	20.00
Final mass of gel / g ± 0.01 g	18.20

(a.i) Calculate the heat energy absorbed by the water, in J. Use sections 1 and 2 of the data booklet. [1]

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(a.ii) Calculate the percentage uncertainty of your answer in (a)(i). [2]

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(b.i) Calculate the enthalpy of combustion of propan-1-ol, in kJ mol^{-1}, stating **one** assumption. If you did not obtain an answer to (a)(i), use 30 000 J, though this is not the correct value. [3]

Calculation: . . . .

Assumption: . . . .

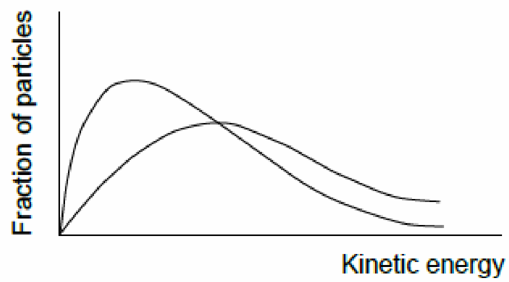
(c) The student also investigates a second gel containing ethanol, with the same percentage by mass as the propan-1-ol in the first gel.

Determine whether the second gel would release more or less energy per gram than the first gel. Use section 14 of the data booklet. [3]

7. Hydrogen is manufactured from methane by a process called steam reforming:

$$CH_4\,(g) + H_2O\,(g) \quad CO\,(g) + 3H_2\,(g) \qquad \Delta H = +206\ \text{kJ mol}^{-1}$$

(a) Explain why the reaction rate increases with temperature, adding annotations to the following Maxwell–Boltzmann graph to assist your explanation. [3]



Explanation: . . . .

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(b) Annotate this Maxwell–Boltzmann distribution graph in (a) to show the effect of a catalyst. [1]

8. An organic compound, **A**, has the following composition by mass when its only combustion products, carbon dioxide and water, are analysed.

C / %	**H / %**
71.93	12.10

(a) Outline why this compound is **not** a hydrocarbon. [1]

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(b) Determine the empirical formula of **A**. [2]

(c) A sample of the vapour of **A** at 200.0 °C, and 1.00×10^5 Pa, has a density of 2.544×10^3 g m^{-3}.

Determine the molar mass and the molecular formula of **A**. [2]

9. Outline why the ionic radius of the phosphide ion, P^{3-}, is greater than that of the sulfide ion, S^{2-}. [1]

10. Sulfur trioxide is an important compound in industry.

(a) Sulfur trioxide has more than one possible Lewis (electron dot) structure.

(a.i) Sketch a Lewis (electron dot) structure for SO_3 which obeys the octet rule. [1]

(a.ii) There is no question aii

(a.iii) State the molecular geometry and the O-S-O bond angle in SO_3. [2]

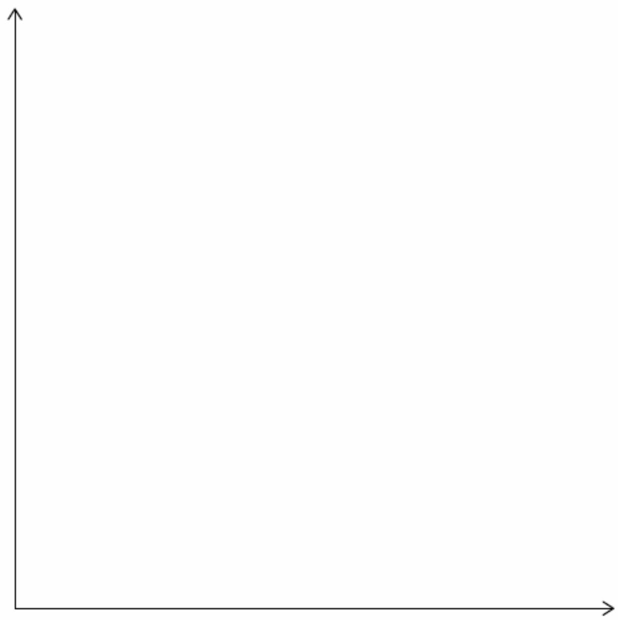
Molecular geometry: . . . .

Bond angle: . . . .

(b) SO_3 (g) is made using the contact process:

$$2SO_2(g) + O_2(g) \quad 2SO_3(g) \qquad \Delta H^{\ominus} < 0$$

(b.i) Sketch a potential energy profile for this reaction on the axes provided. Label E_a and include labels on the axes. [3]



(b.ii) Vanadium pentoxide, V_2O_5, is used as a catalyst. Explain how a catalyst increases the rate of a reaction. [2]

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(b.iii) During the reaction, V_2O_5 changes to V_2O_4. Identify the oxidation states of vanadium in each compound. [1]

V_2O_5: . . . .

V_2O_4: . . . .

11. Draw the skeletal formula of 5-chloro-2-methylpentan-2-ol. [1]

12. Chlorine reacts with methane.

(a(i)) State two features showing that propane and butane are members of the same homologous series. [2]

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(a(ii)) Calculate the enthalpy change of the reaction, ΔH, using section 11 of the data booklet. [3]

$$CH_4\,(g) + Cl_2\,(g) \rightarrow CH_3Cl\,(g) + HCl\,(g)$$

13. State the electron configuration of Chromium, Cr. [1]

14. Ethanol reacts with oleic acid to produce ethyl oleate.

$$C_{17}H_{33}COOH\,(l) + CH_3CH_2OH\,(l) \quad C_{17}H_{33}COOCH_2CH_3\,(l) + H_2O\,(l)$$

Calculate the atom economy of the reaction. [1]

End of Paper 2